New Mangalore Port Authority

Visit Briefing — Questions, Data Asks & Project Status

PROJECT
Maritime Port Intelligence Platform (MPIP) — NMPA use case

PURPOSE OF THIS VISIT
Validate our cost, tariff and vessel-data assumptions directly with NMPA, and identify which of our project's remaining gaps NMPA can realistically help close.

HOW TO USE THIS DOCUMENT
One page per question. Each page gives the background, what we already know/have built, and a boxed line to literally ask on the day. The last page is a one-page cheat sheet for the room.

This briefing was prepared from an internal audit of our own codebase and datasets — every claim below about what is “real” versus “synthetic” in our platform, and every number quoted from the NMPA Scale of Rates (SOR effective 01.05.2026–30.04.2027), was checked line-by-line against source before being written down here.

QUESTION 1 OF 4

Their business runs on tariffs — how does that affect our project?

What NMPA's business actually is

NMPA is a Major Port Authority operating entirely on a government-regulated Scale of Rates (SOR) — currently the 2026-27 SOR, effective 01 May 2026 to 30 Apr 2027. It covers vessel-related charges (port dues, berth hire, pilotage, anchorage, tug hire, shifting) and cargo-related charges (wharfage across 60+ commodity lines, container box rates, transit storage). Rates are ceiling levels set under the Tariff Authority for Major Ports (TAMP) framework and re-indexed every 1 May to 60–100% of the Wholesale Price Index (WPI).

Why this matters directly to us

Our Port Cost Estimator is the one module that actually touches NMPA's real commercial business. Our forecasting and analytics screens are advisory; a cost estimate is something an agent or shipper could act on. Accuracy here isn't optional in the way it might be elsewhere in the platform.

What we've already built

- We transcribed the published SOR directly into our cost engine — wharfage, berth hire, port dues, pilotage, anchorage, tug hire, shifting, and transit storage.
- Example checks that match the SOR line-for-line: crude oil via SPM wharfage ■34.89/MT (coastal) / ■34.90/MT (foreign); dry-bulk berth hire ■0.07602/GRT/hr (coastal), subject to a ■262.83/hr minimum.

Where we're still exposed

- **Static table, manual refresh.** The SOR re-indexes every 1 May; our numbers go stale unless someone manually re-transcribes the new PDF each year.
- **Some components have no fixed numeric rate** in the SOR itself (e.g. cruise-vessel composite charges, container scanning, LDB mandatory user charges) — it just says “as per separate order,” so we can't compute those without that order.
- **Zero real invoices to check against.** Every number is transcribed from the tariff, not verified against an actual bill.

ASK THEM AT THE VISIT

• Do you keep the SOR (or a subset) in a structured/spreadsheet form, or only as the published PDF?
• Could someone in billing or operations sanity-check 3–4 of our sample cost estimates against a real invoice?
• Who owns the annual WPI re-indexation each 1 May — is there a notice we can subscribe to?

QUESTION 2 OF 4

Which APIs do we actually need — and which can NMPA realistically give us?

Not every gap in our platform is NMPA's to fill. The useful exercise is separating data NMPA generates and owns from data NMPA merely consumes from someone else — we should only be asking for the former.

Likely theirs to give — worth asking

- **Vessel-call register** (arrival / berth allotted / departure per ship, IMO number, cargo handled) — HIGH VALUE. Almost certainly exists internally for Port Control / berth planning, even if never published. This single feed would let us build real vessel-stay and berth-occupancy predictions instead of the assumption-based proxy we run today (see Q5b).
- **Live / periodic berth occupancy status** — medium value, likely already exists in their berth planning system.
- **Structured tariff feed** — low technical value but removes our manual PDF-to-code transcription step (see Q1).
- **Circulars & Trade Notice feed** — we already scrape their public “Circular and Trade Notice” page; worth asking if an RSS/API/bulk export exists instead of HTML scraping.

Probably NOT theirs to give — wrong department

- **Cargo manifest / import-export documentation** — that's Customs (ICEGATE) data, not the Port Authority's.
- **Live global AIS / vessel positions at sea** — that's a commercial vendor's product (MarineTraffic, Spire). NMPA's own in-harbour tracking is a related but separate ask — see Q3.
- **Commodity market prices, trade-lane statistics** — government trade-statistics or market-data agencies (DGCIS, World Bank), not the port itself.

ASK THEM AT THE VISIT

- *Does Port Control keep a structured vessel-call log we could get periodic exports from?*
- *Is there an RSS/API for the Circular & Trade Notice page, or should we keep scraping it?*
- *Who's the right contact for each of the above — Traffic, IT, or Harbour Master's office?*

QUESTION 3 OF 4

We're missing live AIS — what should we actually ask for?

Where we are today

Our live-tracking screen is still a fixed, scripted fleet list, not real AIS. But we no longer rely on looking vessels up by hand: we built a real historical archive from NMPA's own public Daily Vessel Position reports (Traffic Department “Proceedings of the Operation Meeting”) and used it to compute the platform's first genuinely real vessel-stay and berth-occupancy numbers.

How we built the historical archive

NMPA's public “Daily Vessel Position” page only displays that day's report, with no visible archive or bulk-download option. Its PDF link, however, follows a predictable, date-based path — and we confirmed the same path resolves for past dates too, meaning a year of history was already sitting on NMPA's own server, simply not linked from the page itself. We built a collector that fetches each date directly and recovered roughly a year of daily reports, which we've since parsed into structured records (arrivals, berthings, sailings, anchorage waits with stated reasons) to link each vessel's actual stay duration. Coverage isn't complete — about 76% of calendar days resolved, closer to 88% of working days once Sundays (which the report doesn't appear to cover) are accounted for — so this is a self-built stopgap, not a substitute for sanctioned access.

Why we still shouldn't just buy commercial AIS

MarineTraffic/Spire-grade satellite AIS is a paid subscription priced for fleet operators, and most of what it would show us — ships already inside NMPA's harbour limits — is redundant with data the port already collects for its own safety and berth-allocation operations via its Vessel Traffic Management System (VTMS) / Port Control, and largely redundant with what our own archive now already recovers from NMPA's public reports.

The actual ask, reframed

Not “give us AIS,” but: can Port Control / VTMS export the vessel-call log it already keeps directly — vessel name / IMO number, ETA/ATA, berth assigned, ATB, ATD, cargo type and quantity handled? That would both fill the ~12–24% gap in our self-built archive and remove our dependency on scraping a page never designed for bulk historical access.

Secondary ask: historical depth

2–3 years of vessel-call history (redacted if needed) would extend what we've already built and let us validate it further against NMPA's own committed performance standard — 37-hour average vessel turnaround, 7-hour average pre-berthing time (SOR Annexure-II). Our own measured median so far, from the self-built archive, is already close to that standard — worth showing them directly.

ASK THEM AT THE VISIT

- *Does Port Control / VTMS maintain a per-vessel call log we could get official bulk access to?*
- *Could a historical extract predating our archive (pre August 2025) be shared?*
- *Who's the right department to ask — Traffic, IT, or Harbour Master's office?*

QUESTION 4 OF 4

Beyond an API, what else could NMPA help with?

Validation, not just data

Asking someone in operations or billing to eyeball a handful of our cost estimates and flag anything that looks off is cheaper than any dataset, and directly de-risks the one module that touches their real business (Q1).

Domain review of our assumptions

Our what-if scenario simulator (vessel-arrival shift %, demand shift %, weather-delay days) currently runs on assumed elasticities, not fitted ones — we say so explicitly in our own documentation. Twenty minutes with a traffic/planning officer sanity-checking whether those assumed numbers are in the right ballpark would materially improve credibility even without any new data.

PCS1x / Port Community System

India's major ports increasingly route cargo and vessel data through the unified Port Community System (PCS1x) rather than bespoke exports. Worth asking directly whether NMPA participates and whether a research/student integration is possible — this could be a single doorway to several of the Q2/Q3 asks instead of separate one-off requests.

A named point of contact

For the annual tariff re-indexation (Q1), and for who to notify if our platform is ever demoed internally and someone spots an error — an ongoing relationship is worth more than a one-time data dump.

Digital / “Smart Port” initiatives

Ask if there's a digital transformation or Smart Port cell (many major ports have one under the Sagarmala programme) — that's the natural home for a student project like ours and the fastest route to a yes on any of the above.

ASK THEM AT THE VISIT

- *Could ops/billing spend a few minutes sanity-checking our cost estimates against a real invoice?*
- *Does NMPA participate in PCS1x, and is a student/research integration possible?*
- *Is there a digital transformation / Smart Port cell we should be talking to?*

PROJECT STATUS — PART A

What in our platform is genuinely real today?

Before asking NMPA for anything, we audited our own platform end-to-end. This page and the next are an honest summary — useful both for us and as a credibility signal if NMPA staff ask “what exactly are you showing me.”

Cargo forecasting — REAL, validated

RandomForest model trained on 1,354 real monthly NMPA commodity-traffic records (2021–2026), backtested out-of-sample with measured accuracy (WAPE ≈ 17%, ≈ 83% accuracy vs. a seasonal-naive baseline).

Vessel-call forecasting & anomaly detection — REAL, partially validated

Same real dataset, RandomForest / IsolationForest models. Anomaly detection is real ML but unsupervised — there's no labeled ground truth to check it against.

Port cost estimator — REAL tariff data, not yet invoice-validated

Built directly from the published SOR (see Q1) — real numbers, not invented — but never checked against an actual NMPA bill.

Berth / facility routing — REAL NMPA facility data

Uses real berths.csv / berth_capacity.csv NMPA facility data with rule-based commodity-to-berth matching. Our own audit caught and fixed a substring-matching bug that had been silently routing crude oil, LPG, coal and containers to Berth 1 (a shallow general-cargo berth) because “Berth 1” matched as a substring of “Berth 10” through “Berth 17.”

Vessel stay duration & real call spacing — REAL, self-built from NMPA's own reports

See Q3: roughly a year of NMPA's public Daily Vessel Position reports collected and parsed, vessel identities resolved across reporting inconsistencies, and each call's berthing-to-sailing duration linked from real timestamps where available. High-confidence median stay duration measured so far is within a few hours of NMPA's own 37-hour turnaround standard (SOR Annexure-II).

NMPA Circulars & Trade Notices — not built, but confirmed easy

Confirmed scrapable from the public “Circular and Trade Notice” page with no NMPA dependency — a fully achievable addition using only public data.

PROJECT STATUS — PART B

What we can't solve yet, and the specific reason why

Vessel stay duration / berth occupancy — PARTIALLY SOLVED, self-built

No longer blocked: see Project Status Part A. What's missing is completeness, not the mechanism — about 12-24% of days aren't recoverable from the public page alone, and roughly a quarter of linked calls have no confirmed departure event yet (vessel may still be in port, or its sailing fell in a gap). Official bulk access (Q3) would close both gaps and extend history further back.

Real-time vessel tracking (AIS) — BLOCKED, though partially superseded

True live tracking is still not solved (cost of commercial AIS, redundant with NMPA's own VTMS) — but our self-built archive already recovers most of the value we originally wanted AIS for (real stay/occupancy data), just not live, and not with vessel position at sea.

Trade intelligence (trade lanes, commodity prices, “opportunities”) — BLOCKED, but not on NMPA

We have no origin/destination or market-price dataset at all — these numbers are currently randomly generated on every request. Solvable independently of NMPA, via DGCIS / Ministry of Commerce trade statistics and World Bank commodity price data — an ask for us to pursue on our own, not something to raise at the port visit.

Digital twin, executive KPI dashboard, AI copilot, security/SOC module — NOT YET REAL

Currently fully synthetic — randomly generated on every request, including fabricated revenue figures and a fake “847,203 postgresql_records” value. Not “blocked” so much as never started for real. Needs either a real data source behind each screen, or the UI needs to visibly say “simulated” as clearly as our own code comments already do.

What-if scenario simulator — mechanism real, calibration missing

The simulator itself runs correctly, but its elasticity constants are documented assumptions, not fitted to real data — same root cause as the vessel-stay gap above: no real congestion/delay dataset exists yet to calibrate against.

FOR THE ROOM

One-page cheat sheet — what to literally ask

MUST-ASK

- **1.** “Does Port Control / VTMS keep a per-vessel call log we could get official bulk access to — to replace our own scraped archive and fill its coverage gaps?” *(Q3 — highest impact, extends what we've already self-built.)*
- **2.** “Is there a digital transformation / Smart Port / PCS1x integration point we should be talking to instead of, or in addition to, Traffic/IT?” *(Q4 — could be a single doorway to several asks.)*
- **3.** “Could someone in billing or operations sanity-check a handful of our cost estimates against a real invoice?” *(Q1/Q4 — cheapest, highest-credibility ask.)*

NICE-TO-ASK

- “Is the SOR maintained anywhere in structured/spreadsheet form, or only as the published PDF?” (Q1)
- “Is there an RSS/API for the Circular & Trade Notice page, or should we keep scraping it?” (Q2)
- “Who owns the annual WPI re-indexation announcement each 1 May, and is there a way to subscribe to it?” (Q1)
- “Could a traffic/planning officer spend 20 minutes reviewing our what-if simulator's assumptions for realism?” (Q4)

WHAT NOT TO ASK NMPA FOR — redirect elsewhere

- Global AIS / MarineTraffic-grade live positions — a commercial vendor's product, not the port's.
- Trade-lane and commodity-price data — DGCIS / World Bank, not the port.
- Import-export cargo manifests — Customs (ICEGATE), not the Port Authority.

Prepared from an internal, line-by-line audit of the project codebase and the NMPA Scale of Rates (effective 01.05.2026–30.04.2027). Every “real” / “synthetic” claim on the Project Status pages traces to a specific function or dataset in the codebase.